Business Cycles

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Advanced Macroeconomics – Part 2 – Lecture 1

November 2026

Roadmap

- Introduction: a (very) brief history of business cycle macro
- Business cycle facts: US and Europe
- The RBC model as a benchmark
- Business Cycle Accounting (Chari, Kehoe, McGrattan, 2007)
  - Prototype economy and the four wedges
  - Equivalence results: from frictions to wedges
  - Accounting procedure
- Numerical methods: log-linearization
- Results: postwar US 1948–2026, Great Recession and COVID-19, Great Recession in Europe

Introduction:
a brief history of business cycle macro

A brief history of business cycle macro

Poor attempt to fit complex history into a simple narrative …
Some key models and ideas:

- 1920s–1940s: measurement of cycles (NBER) and impulse–propagation models
- 1930s–1960s: Keynesian model and large-scale macroeconometric models
- 1970s: Lucas critique, microfoundations and rational expectations
- 1980s: Real Business Cycle (RBC) model
- 1990s–2010s: New Keynesian DSGE models: *many shocks, many frictions*
- 2007: Business Cycle Accounting: which frictions matter? ← **today**
- Since the 2010s: heterogeneity, HANK ← *rest of this course*

1920s–1960s: Measurement, impulses, and Keynes

- Business cycles first a measurement problem
  - Mitchell (1927), Burns & Mitchell (1946) *"Measuring Business Cycles"* at the NBER
  - cycle = expansions and contractions occurring at about the same time in *many* economic activities; recurrent but not periodic, lasting from more than one year to ten or twelve years
  - still today: NBER and CEPR (euro area) business cycle dating committees
- Frisch (1933), Slutsky (1927): impulse vs. propagation
  - random shocks fed into a stable linear dynamic system generate cycles
  - still the backbone of modern DSGE models: $X_{t+1} = AX_t + B\varepsilon_{t+1}$
- Keynes (1936): "General Theory", Hicks (1937): IS–LM
  - persistent involuntary unemployment: output determined by aggregate demand
  - scope for countercyclical fiscal policy: high MPC $\Rightarrow$ high multipliers
- 1950s–60s: large-scale macroeconometric models without microfoundations (Tinbergen (1936-39), Klein & Goldberger (1955), Cowles commission…)
  - consumption function $C = \alpha + \gamma(Y - T)$, Phillips curve …estimated equation by equation
  - used for forecast and policy counterfactuals

1970s: Lucas critique and rational expectations

- Stagflation: breakdown of the (exploitable) Phillips curve – Friedman (1968), Phelps (1968)
- Lucas (1976) "Econometric Policy Evaluation: A Critique"
  - parameters of macroeconometric models are not policy-invariant: behavior changes with the rules of the game
  - ⇒ build models from structural (policy-invariant) primitives: preferences, technology, resource constraints
- Rational (= model-consistent) expectations: $p_t^e = \mathbb{E}_{t-1}(p_t)$
  Muth (1961), Lucas (1972), Sargent & Wallace (1975)
- Lucas (1977) "Understanding Business Cycles"
  - cycles = comovements of deviations from trend of many aggregate series
    "w.r.t. qualitative behaviors of co-movements among series, business cycles are all alike"
  - ⇒ look for a single *general equilibrium* explanation of a set of stylized facts
- Sims (1980) "Macroeconomics and Reality": VARs, few identifying restrictions

1980s–2000s: RBC, New Keynesian DSGE, and BCA

- Kydland & Prescott (1982), Long & Plosser (1983): Real Business Cycle model
  - stochastic Neoclassical Growth model (NCG) + TFP shocks, *calibration*
  - Prescott (1986) "Theory Ahead of Business Cycle Measurement": TFP shocks account for a large share of US fluctuations
  - Summers (1986) "Some Skeptical Observations": what are these shocks? Technology regress?
- 1990s–2000s: New Keynesian DSGE: RBC core + nominal rigidities + many frictions (habits, adjustment costs, capacity utilization, financial frictions) + many shocks
  - Four agenda coming together: (i) Nairu/labor-market work, (ii) empirical business-cycle, (iii) rules and determinacy, (iv) central banks testing policy rules
  - Woodford (2003), Gali (2008), Bernanke, Gertler & Gilchrist (1999), Christiano, Eichenbaum & Evans (2005), Smets & Wouters (2007)

$\Rightarrow$ ***Question:*** among these frictions, which ones matter quantitatively for business cycles?

- Chari, Kehoe & McGrattan (2007): Business Cycle Accounting
  - use RBC model as *measurement device*: deviations from optimality conditions = wedges
  - guide researchers towards promising classes of models

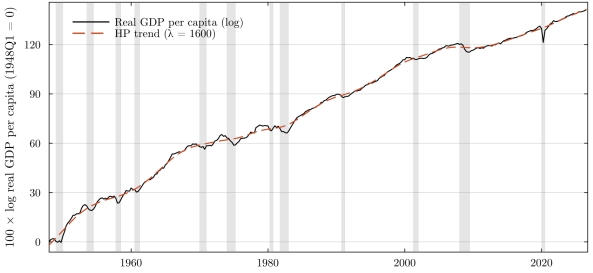
Business cycle facts

Trends and cycles

- Decompose (log) aggregate series into trend and cycle: $\log y_t = \bar{y}_t + \hat{y}_t$
- Hodrick–Prescott (1997) filter: trend solves

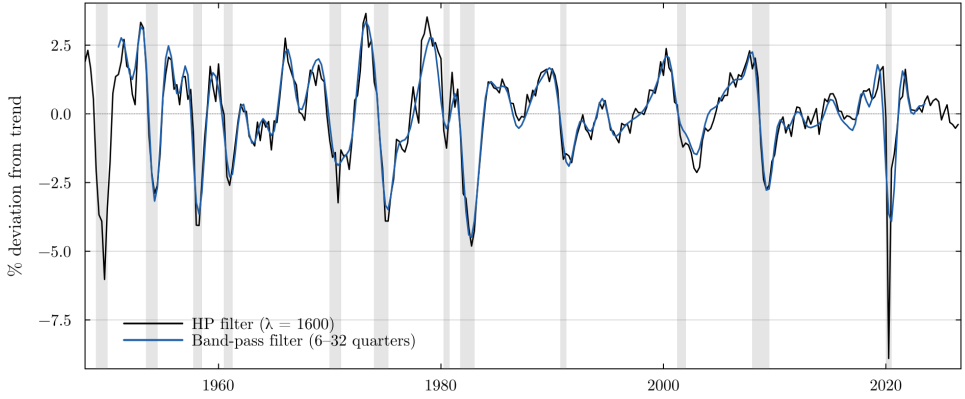
$$\min_{\{\bar{y}_t\}} \sum_t (\log y_t - \bar{y}_t)^2 + \lambda \sum_t \left[(\bar{y}_{t+1} - \bar{y}_t) - (\bar{y}_t - \bar{y}_{t-1})\right]^2, \qquad \lambda = 1600 \text{ (quarterly)}$$

- Band-pass filter (Baxter & King, 1999): keep periodicities between 6 and 32 quarters



Log real GDP per capita (US, 1948Q1–2026Q3) and its HP trend ($\lambda = 1600$). NBER recessions shaded

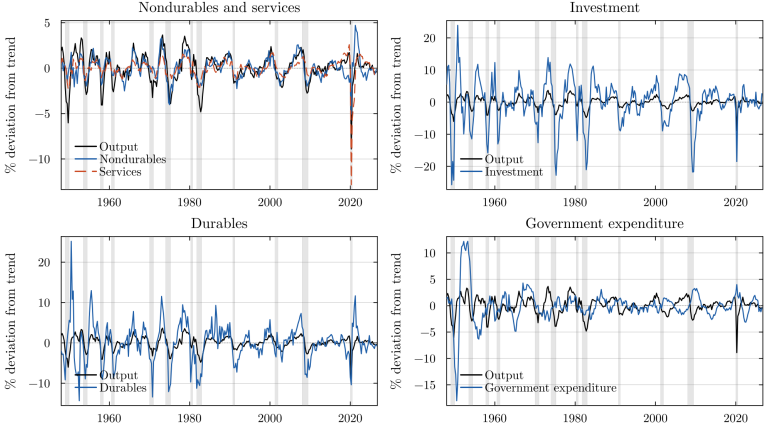
GDP de-trended



Cyclical component of US real GDP per capita, in %: HP filter ($\lambda = 1600$) vs. band-pass (6–32 quarters)

- Fluctuations of ± 2–4% around trend; recessions = sharp and short, expansions = long

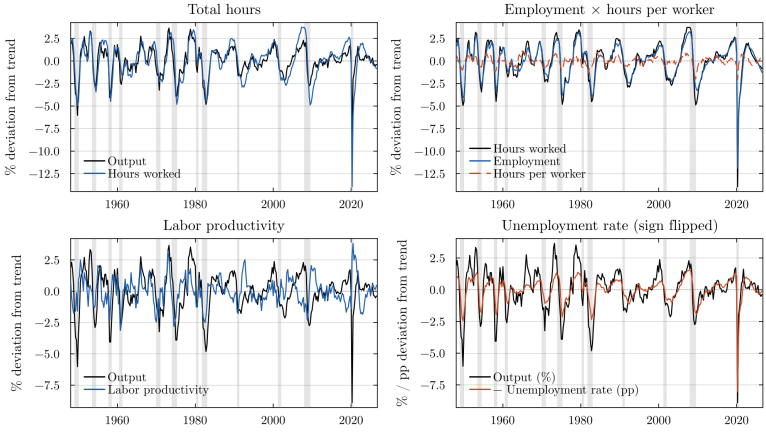
Stylized facts – Consumption and investment



Cyclical components (HP, %) of output vs. consumption, investment and government expenditure, US per capita

- Consumption (nondurables, services) less volatile than output, durables and investment much more volatile
- All strongly procyclical – except government expenditure, roughly acyclical (war build-up)

Stylized facts – Hours and productivity



Cyclical components (HP) of output vs. hours, labor productivity, and unemployment (pp, sign flipped)

- Hours as volatile as output, mostly through employment (extensive margin), slightly lagging
- Labor productivity mildly procyclical; unemployment strongly countercyclical (Okun's law)

Stylized facts – US business cycle moments

Table: US data, 1948Q1–2019Q4, HP-filtered ($\lambda = 1600$), quarterly, per capita

Variable	$\sigma(\cdot)$	$\sigma(\cdot)/\sigma(y)$	$\rho(\cdot, y)$	$\rho(\cdot, h)$	Autocorr.(1)
Output	1.59	–	–	0.87	0.85
Consumption	1.22	0.77	0.77	0.68	0.81
Nondurables	1.13	0.71	0.69	0.63	0.80
Services	0.78	0.49	0.73	0.70	0.85
Durables	4.76	2.99	0.60	0.48	0.74
Investment	7.20	4.53	0.86	0.77	0.81
Fixed investment	4.74	2.98	0.81	0.81	0.89
Change in inventories / GDP (pp)	0.43	0.27	0.64	0.48	0.50
Government expenditure	3.19	2.00	0.14	0.03	0.91
Hours worked	1.92	1.20	0.87	–	0.91
Employment	1.60	1.01	0.81	0.97	0.92
Hours per worker	0.51	0.32	0.71	0.70	0.81
Labor productivity	1.07	0.67	0.38	-0.10	0.72
Unemployment rate (pp)	0.81	0.51	-0.87	-0.96	0.91

- Volatility ranking: $\sigma(c) < \sigma(y) \approx \sigma(h) < \sigma(i)$. All series are persistent

Stylized facts – Europe since 1995

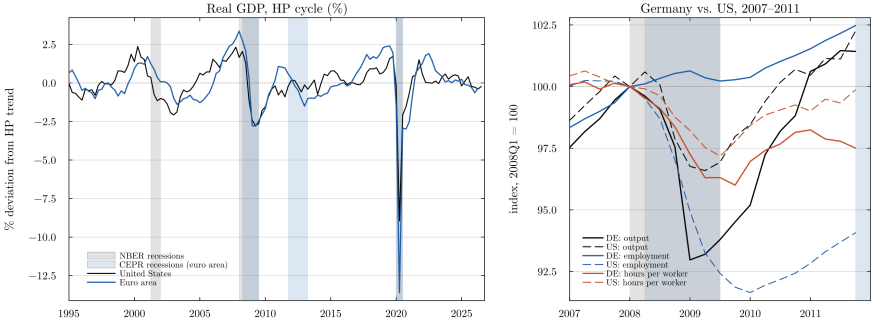
Table: Quarterly, 1995Q1–2019Q4, HP-filtered ($\lambda = 1600$), log real aggregates

	$\sigma(y)$	$\frac{\sigma(c)}{\sigma(y)}$	$\frac{\sigma(i)}{\sigma(y)}$	$\frac{\sigma(g)}{\sigma(y)}$	$\frac{\sigma(n)}{\sigma(y)}$	$\frac{\sigma(h)}{\sigma(y)}$	$\rho(c,y)$	$\rho(i,y)$	$\rho(n,y)$
United States	1.03	0.79	4.25	1.21	1.37	1.71	0.87	0.91	0.83
Euro area	1.12	0.61	2.49	0.47	0.53	0.72	0.81	0.83	0.80
Germany	1.44	0.41	2.14	0.61	0.41	0.61	0.48	0.88	0.46
France†	0.95	0.76	2.51	0.48	0.54	0.77	0.65	0.90	0.28
Italy	1.23	0.81	2.10	0.90	0.48	0.78	0.73	0.87	0.58
Spain	1.21	1.19	2.89	0.95	1.22	1.25	0.88	0.82	0.94
United Kingdom	1.10	0.91	–	–	0.44	–	0.83	–	0.74

Aggregates (not per capita). EA: GDP EA19, components and labor EA20. c = household consumption, i = GFCF (EA includes Irish IP investment), g = gov. consumption (US: incl. gov. investment). n = employment, h = total hours (US: nonfarm business). †France: n from 2003 (OECD). UK: Eurostat stops in 2020, partial data.

- Same qualitative facts: $\sigma(c) < \sigma(y) < \sigma(i)$, c, i, n procyclical. Investment "only" 2-3× output (US: 4×)
- Employment adjusts much less than in the US ($\sigma(n)/\sigma(y) \approx 0.4$–$0.5$ vs. 1.4), *except Spain* (temp. contracts)

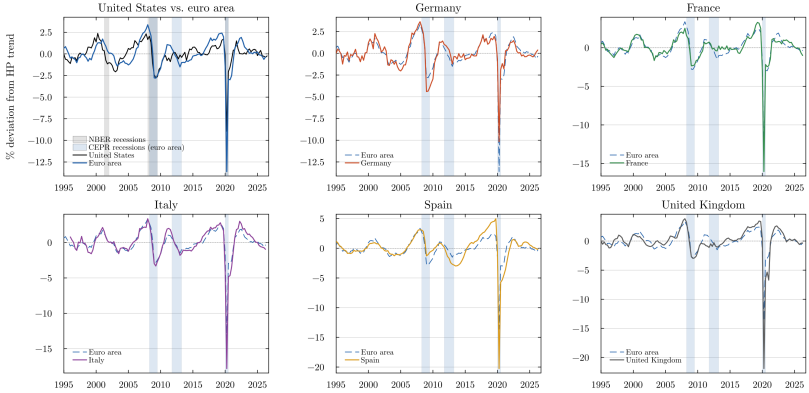
Stylized facts – Euro area vs. US since 1995



Left: HP cycle of real GDP, NBER / CEPR recessions. Right: Germany (solid) vs. US (dashed), 2008Q1 = 100

- Business cycles highly synchronized across the Atlantic (Great Recession, Covid-19): $\rho(y^{US}, y^{EA}) = 0.81$
- . . . but the euro area has a double-dip: sovereign debt crisis 2011–2013
- Labor adjustment differs: *intensive* margin in Europe (short-time work, *Kurzarbeit*): the German "labor market miracle" in 2009 vs. large employment losses in the US

Euro area business cycles, country by country



HP cycle of real GDP (%), each country vs. the euro area (dashed), 1995–2026. CEPR euro-area recessions shaded

- Strong co-movement: correlation with the euro-area cycle 0.92–0.96 for DE, FR, IT, ES, UK
- Amplitudes differ: $\sigma(y)$ larger in Spain & Italy; Germany hit hardest in 2009 (exports), Italy and Spain in 2011–13 (sovereign debt crisis)

Business cycle facts – Summary

- Business cycles = comovements: output, consumption, investment, hours, productivity move together
- Volatility:
  - investment $\approx$ 3–4$\times$ output; consumption (nondurables + services) $\approx 0.5\times$ output
  - hours $\approx$ output, mostly through employment in the US
- Persistence: first-order autocorrelation $\approx$ 0.8–0.9
- Labor productivity mildly procyclical, almost uncorrelated with hours
- Same broad picture in Europe, with differences in labor market adjustment

$\Rightarrow$ Can a *simple* general-equilibrium model reproduce these facts?

The RBC model as a benchmark

The RBC model as a benchmark

- Neoclassical growth model did a good job at explaining *growth* facts
  Kydland & Prescott (1982): feed in the *fluctuations* of the Solow residual as well

- Representative household and firm, complete markets:

$$\max \ \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \, U(c_t, l_t)$$

$$c_t + k_{t+1} = A_t \, k_t^{\alpha} \, l_t^{1-\alpha} + (1-\delta) k_t, \qquad \log A_{t+1} = \rho \log A_t + \varepsilon_{t+1}$$

- Welfare theorems hold $\Rightarrow$ solve the planner's problem $V(k, A)$

- Calibration (quarterly, King & Rebelo, 1999): $U = \log c + \psi \log(1-l)$, $\alpha = 0.33$, $\delta = 0.016$, $\beta = 0.992$, $\bar{l} = 1/3$, $G/Y = 0.20$
  Solow residual $\log A_t = \log y_t - \alpha \log k_t - (1-\alpha) \log l_t$, linearly detrended (FRED, 1948–2019, k_t by perpetual inventory) $\Rightarrow \rho = 0.975$, $\sigma_\varepsilon = 0.0065$

RBC vs. data: second moments

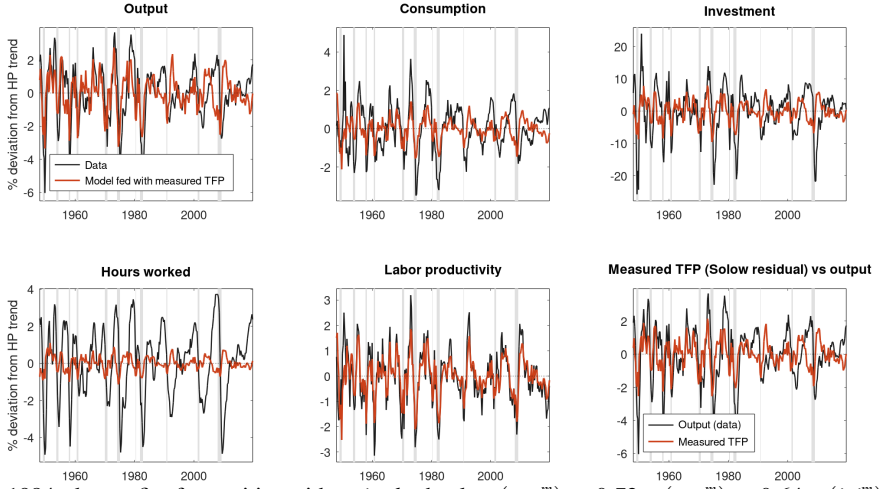
Table: US data vs. RBC with measured TFP, HP-filtered ($\lambda = 1600$), 1948Q1–2019Q4

	$\sigma(\cdot)/\sigma(y)$			$\rho(\cdot, y)$			$\rho(\cdot, h)$		
Variable	Data	RBC	Indiv.	Data	RBC	Indiv.	Data	RBC	Indiv.
Output: $\sigma(y)$, %	1.59	1.10	1.37	–	–	–	0.87	0.96	0.97
Consumption	0.77	0.53	0.47	0.77	0.97	0.95	0.68	0.87	0.84
Investment	4.53	2.89	3.05	0.86	0.99	0.99	0.77	0.99	0.99
Hours worked	1.20	0.34	0.58	0.87	0.96	0.97	–	–	–
Labor productivity	0.67	0.68	0.47	0.38	0.99	0.95	-0.10	0.92	0.84

RBC: $\log(1-l)$ utility, Frisch elasticity $(1-\bar{l})/\bar{l} = 2$. Indiv.: $U = \log c - \psi l$ (Hansen 1985, Rogerson 1988)

- Measured TFP alone: $\sigma(y^m)/\sigma(y) = 0.69 \Rightarrow$ 47% of output variance (74% with indivisible labor)
- Right ranking $\sigma(c) < \sigma(y) < \sigma(i)$, strong comovement. But: hours too smooth, productivity too procyclical (vs. $\rho(y/h, h) \approx 0$ in the data), weak propagation: autocorr.(1) of y is 0.72 vs. 0.85

RBC fed with the measured Solow residual



- 1948–1984: decent fit of quantities with a *single* shock: $\rho(y, y^m) = 0.72$, $\rho(c, c^m) = 0.64$, $\rho(i, i^m) = 0.61$ …but hours: $\rho(h, h^m) = 0.14$!
- 1985–2019: Solow residual becomes acyclical, $\rho(\log A, y) = -0.03 \Rightarrow \rho(y, y^m) = -0.03$, $\rho(h, h^m) = -0.58$

RBC: discussion

- If the model is right, fluctuations are efficient: no scope for stabilization policy
- TFP is a black box: do people "forget" technologies in recessions?
- The Solow residual is contaminated by unmeasured utilization and labor hoarding (Basu, Fernald & Kimball, 2006): utilization-adjusted TFP is roughly acyclical, and technology improvements lower hours on impact

$\Rightarrow$ Rather than *one shock*, let the data tell us *which margins* of the model fail: wedges

Roadmap

- Introduction: a (very) brief history of business cycle macro
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- The RBC model as a benchmark
- **Business Cycle Accounting (Chari, Kehoe, McGrattan, 2007)**
  - **Prototype economy and the four wedges**
  - **Equivalence results: from frictions to wedges**
  - **Accounting procedure**
- Numerical methods: log-linearization
- Results: postwar US 1948–2026, Great Recession and COVID-19, Great Recession in Europe

Business Cycle Accounting

Business Cycle Accounting: the idea

- Researchers building quantitative models face hard choices: where to introduce frictions?

- Method with two components:

  1. ***Equivalence result***: a large class of models with frictions is equivalent to a prototype growth model with time-varying wedges that look like
     - productivity: efficiency wedge A_t
     - labor income tax: labor wedge $1 - \tau_{lt}$
     - investment tax: investment wedge $1/(1 + \tau_{xt})$
     - government consumption: government consumption wedge g_t

  2. ***Accounting procedure***:
     - *measure* the wedges using data + equilibrium conditions of the prototype model
     - *feed* them back into the model, one at a time and in combinations
     - how much of output, labor, investment fluctuations does each wedge account for?

The prototype economy

- Events s_t, histories $s^t = (s_0, \ldots, s_t)$, probabilities $\pi_t(s^t)$. Wedges are functions of s^t
- Households maximize, with population $N_t = (1+\gamma_n)^t$:

$$\max \sum_{t=0}^{\infty} \sum_{s^t} \beta^t \pi_t(s^t)\, U\big(c_t(s^t), l_t(s^t)\big) N_t$$

$$\text{s.t.} \quad c_t + \big[1+\tau_{xt}\big]\, x_t = \big[1-\tau_{lt}\big]\, w_t l_t + r_t k_t + T_t$$

$$(1+\gamma_n)\, k_{t+1} = (1-\delta) k_t + x_t$$

- Firms maximize profits with labor-augmenting technical progress $(1+\gamma)^t$:

$$\max_{k_t, l_t} A_t\, F\big(k_t, (1+\gamma)^t l_t\big) - r_t k_t - w_t l_t$$

- Government: $g_t + T_t = \tau_{lt} w_t l_t + \tau_{xt} x_t$ (distortionary "tax" revenues rebated lump sum)

Equilibrium conditions and the four wedges

- Equilibrium summarized by four conditions – one wedge in each:

Resource constraint	$c_t + x_t + g_t = y_t$
Production function	$y_t = A_t F\left(k_t, (1+\gamma)^t l_t\right)$
Intratemporal (labor)	$-\dfrac{U_{lt}}{U_{ct}} = \left[1 - \tau_{lt}\right] A_t (1+\gamma)^t F_{lt}$
Intertemporal (Euler)	$U_{ct}\left[1 + \tau_{xt}\right] = \beta\, \mathbb{E}_t\left[U_{ct+1}\left\{A_{t+1}F_{kt+1} + (1-\delta)\left[1 + \tau_{xt+1}\right]\right\}\right]$

- Each wedge = the ***overall*** distortion to the relevant equilibrium condition
  - labor wedge = distortions to labor supply (households) *and* labor demand (firms)
  - investment wedge = distortions to household *and* firm Euler equations
- With 4 wedges and 4 equations: wedges account for 100% of (y_t, l_t, x_t, c_t) by construction

Interpreting the wedges

- Wedges are reduced forms for many frictions in "detailed" economies:

Wedge	**Examples of frictions / shocks in detailed models**
Efficiency A_t	technology, input-financing frictions, misallocation, work rules, mismeasurement, disorganization
Labor $1 - \tau_{lt}$	sticky wages + monetary shocks, sticky prices (markups), unions and cartels (Cole & Ohanian, 2004), search frictions, taxes
Investment $1/(1 + \tau_{xt})$	credit-market frictions: Bernanke, Gertler & Gilchrist (1999), Carlstrom & Fuerst (1997), intertemporal preference shocks
Gov. consumption g_t	government spending, net exports (open economy)

- Many models map into the *same* configuration of wedges
  $\Rightarrow$ BCA identifies a class of promising models, not one model or one primitive shock

Equivalence I: Input-financing frictions → efficiency wedge

- Detailed economy: two sectors $i = 1, 2$, gross output $q_t = q_{1t}^{\phi} q_{2t}^{1-\phi}$, $q_{it} = m_{it}^{\theta} z_{it}^{1-\theta}$
  - intermediate inputs m_{it} must be financed in advance at rate $R_{it} = 1 + \tau_{it}$
  - sector 1 more financially constrained (small firms): $\tau_{1t} > \tau_{2t}$
  - value added $z_{1t} + z_{2t} = F(k_t, l_t)$
- ***Proposition 1***: the aggregate allocations coincide with a prototype economy with

$$A_t = \kappa \left(a_{1t}^{1-\phi} a_{2t}^{\phi}\right)^{\frac{\theta}{1-\theta}} \left[1 - \theta(a_{1t} + a_{2t})\right], \qquad a_{1t} = \frac{\phi}{1+\tau_{1t}}, \;\; a_{2t} = \frac{1-\phi}{1+\tau_{2t}}$$

$$\frac{1}{1-\tau_{lt}} = \frac{1}{1-\theta}\left[1 - \theta(a_{1t} + a_{2t})\right], \qquad \tau_{kt} = \tau_{lt}$$

- If the *average* spread $a_{1t} + a_{2t}$ is constant but *relative* spreads fluctuate: only A_t moves ⇒ financial frictions look like technological regress!

Equivalence II: Sticky wages → labor wedge

- Detailed monetary economy (Bordo, Erceg & Evans, 2000):
  - unions $j \in [0,1]$ set nominal wages $W_{(j,s^{t-1})}$ *one period in advance*, face labor demand $l^d_{(j,s^t)} = \left(W_{(s^{t-1})}/W_{(j,s^{t-1})}\right)^{\frac{1}{1-\nu}} l_{(s^t)}$
  - money supply shocks $M_{(s^t)} = \mu_{(s^t)} M_{(s^{t-1})}$; steady state: $\frac{W}{P} = \frac{1}{\nu}\left(-\frac{U_l}{U_c}\right)$
- ***Proposition 2***: equilibrium allocations and prices coincide with a *real* prototype economy with labor wedge

$$1 - \tau_l(s^t) = -\frac{U^*_l(s^t)}{U^*_c(s^t)} \frac{1}{F^*_l(s^t)}$$

- Nominal wages pre-set: a monetary shock moves P_t but not $W_{t-1} \Rightarrow$ real wage, hence the gap between MRS and MPL, fluctuates $\Rightarrow$ measured as a labor wedge
- Similarly (CKM appendix):
  - BGG (1999) agency costs → investment wedge (tax on capital income)
  - Open economy borrowing/lending → net exports = government consumption wedge

The accounting procedure

- ***Markovian implementation***: state $s_t = (s_{At}, s_{lt}, s_{xt}, s_{gt})$ = the four wedges, following a vector AR(1)

$$s_{t+1} = P_0 + P s_t + \varepsilon_{t+1}, \qquad \varepsilon_{t+1} \sim \mathcal{N}(0, V), \qquad V = QQ'$$

  i.e. agents use past wedges to forecast future wedges

- ***Step 1***: estimate (P_0, P, V) by maximum likelihood using data on (y_t, l_t, x_t, g_t)
- ***Step 2***: uncover the realized wedges s_t^d such that the model reproduces the data

$$y_t^d = y(s_t^d, k_t), \qquad l_t^d = l(s_t^d, k_t), \qquad x_t^d = x(s_t^d, k_t), \qquad g_t = g_t^d$$

  with $k_{t+1} = (1-\delta)k_t + x_t^d$ (perpetual inventory)

- ***Step 3***: experiments: "efficiency wedge alone" economy: $A_t = s_{At}$, other wedges at constants $\bar{\tau}_l, \bar{\tau}_x, \bar{g}$
  - keep the *same* stochastic process: removes the direct effect of the other wedges, keeps their forecasting effect

Measuring the wedges

- Functional forms: $F(k,l) = k^{\alpha} l^{1-\alpha}$, $U(c,l) = \log c + \psi \log(1-l)$
- Efficiency wedge: Solow residual, from production function

$$A_t = \frac{y_t}{k_t^{\alpha}\left((1+\gamma)^t l_t\right)^{1-\alpha}}$$

- Labor wedge: MRS / MPL, from intratemporal condition

$$1 - \tau_{lt} = \frac{\psi}{1-\alpha} \frac{c_t}{y_t} \frac{l_t}{1-l_t}$$

- Government consumption wedge: directly $g_t = y_t - c_t - x_t$ (incl. net exports)
- Investment wedge: Euler equation involves $\mathbb{E}_t[\cdot]$
  $\Rightarrow$ requires *solving* the model and knowing the stochastic process (Step 1)

Numerical methods: log-linearization

Solving the model: detrending and steady state

- Non-linear system of stochastic difference equations under rational expectations: in general no closed form ⇒ numerical approximation
- Step 0: detrend: $\hat{v}_t = V_t/\left[N_t(1+\gamma)^t\right]$. With $\hat{\beta} = \beta/(1+\gamma)$ (log utility):

$$\hat{c}_t + \hat{g}_t + (1+\gamma)(1+\gamma_n)\hat{k}_{t+1} - (1-\delta)\hat{k}_t = \hat{y}_t = A_t \hat{k}_t^{\alpha} l_t^{1-\alpha}$$
$$\frac{\psi\,\hat{c}_t}{1-l_t} = (1-\tau_{lt})(1-\alpha)\frac{\hat{y}_t}{l_t}$$
$$\frac{1+\tau_{xt}}{\hat{c}_t} = \hat{\beta}\,\mathbb{E}_t\Big[\frac{1}{\hat{c}_{t+1}}\Big(\alpha\frac{\hat{y}_{t+1}}{\hat{k}_{t+1}} + (1-\delta)(1+\tau_{xt+1})\Big)\Big]$$

- Step 1: steady state for constant wedges, e.g.

$$\alpha\frac{\hat{y}}{\hat{k}} = (1+\tau_x)\Big(\frac{1}{\hat{\beta}} - (1-\delta)\Big) \qquad \Rightarrow \qquad \hat{k}/l,\ \ \hat{c},\ \ l,\ \ \hat{y}$$

- Two options: ***global*** non-linear (finite elements) vs. ***local*** log-linear

Log-linearization

- Define log-deviations from steady state: $\widetilde{x}_t = \log x_t - \log \bar{x} \approx \frac{x_t - \bar{x}}{\bar{x}}$.
  First-order Taylor expansion in $\log x$: $f(x_t) \approx f(\bar{x}) + f'(\bar{x})\,\bar{x}\,\widetilde{x}_t$
- Useful rules: $x_t^a y_t^b \to a\widetilde{x}_t + b\widetilde{y}_t, \qquad x_t + y_t = z_t \;\to\; \bar{x}\widetilde{x}_t + \bar{y}\widetilde{y}_t = \bar{z}\widetilde{z}_t$
- Applied to the static conditions (with $\widetilde{\tau}_{lt} := d\log(1-\tau_{lt})$):

$$\begin{aligned}
&\text{Production:} && \widetilde{y}_t = \widetilde{A}_t + \alpha\widetilde{k}_t + (1-\alpha)\widetilde{l}_t \\
&\text{Labor FOC:} && \widetilde{c}_t + \frac{\bar{l}}{1-\bar{l}}\widetilde{l}_t = \widetilde{\tau}_{lt} + \widetilde{y}_t - \widetilde{l}_t \quad \Leftrightarrow \quad \widetilde{\tau}_{lt} = \widetilde{c}_t - \widetilde{y}_t + \frac{1}{1-\bar{l}}\widetilde{l}_t \\
&\text{Resources:} && \bar{y}\widetilde{y}_t = \bar{c}\widetilde{c}_t + \bar{x}\widetilde{x}_t + \bar{g}\widetilde{g}_t \\
&\text{Capital:} && (1+\gamma)(1+\gamma_n)\bar{k}\,\widetilde{k}_{t+1} = (1-\delta)\bar{k}\,\widetilde{k}_t + \bar{x}\,\widetilde{x}_t
\end{aligned}$$

$\Rightarrow$ Given data $(\widetilde{y}_t, \widetilde{c}_t, \widetilde{l}_t, \widetilde{k}_t)$, the efficiency and labor wedges are linear residuals

Log-linearizing the Euler equation

- Euler equation, with $\widetilde{\tau}_{xt} := d\log(1+\tau_{xt})$: take logs, totally differentiate, use $\hat{\beta}\alpha\frac{\bar{y}}{\bar{k}} = (1+\tau_x)\big(1-\hat{\beta}(1-\delta)\big)$:

$$\widetilde{\tau}_{xt} - \widetilde{c}_t = \mathbb{E}_t\Big[-\widetilde{c}_{t+1} + \big(1-\hat{\beta}(1-\delta)\big)\big(\widetilde{y}_{t+1} - \widetilde{k}_{t+1}\big) + \hat{\beta}(1-\delta)\,\widetilde{\tau}_{xt+1}\Big]$$

  i.e. the marginal product of capital $\alpha\hat{y}/\hat{k}$ enters through $\widetilde{y}_{t+1} - \widetilde{k}_{t+1}$

- Measuring $\widetilde{\tau}_{xt}$ requires expectations: e.g. with AR(1) $\mathbb{E}_t\widetilde{\tau}_{xt+1} = \rho_x\widetilde{\tau}_{xt}$

$$\big(1-\hat{\beta}(1-\delta)\rho_x\big)\widetilde{\tau}_{xt} = \mathbb{E}_t\Big[\widetilde{c}_t - \widetilde{c}_{t+1} + \big(1-\hat{\beta}(1-\delta)\big)\big(\widetilde{y}_{t+1} - \widetilde{k}_{t+1}\big)\Big]$$

- Investment wedge $\approx$ residual between *expected* consumption growth and *expected* return on capital
  - in CKM: expectations from the model's decision rules + estimated VAR for all wedges
  - observationally equivalent to an intertemporal preference shock

Solving the linear system

- Static conditions $\Rightarrow \widetilde{l}_t, \widetilde{y}_t, \widetilde{c}_t, \widetilde{x}_t$ linear in $(\widetilde{k}_t, \widetilde{k}_{t+1}, s_t)$.
  Plug into the Euler equation:

$$a_0\,\mathbb{E}_t\widetilde{k}_{t+2} + a_1\widetilde{k}_{t+1} + a_2\widetilde{k}_t + b_0\,\mathbb{E}_t s_{t+1} + b_1 s_t = 0$$

- Undetermined coefficients: guess $\widetilde{k}_{t+1} = \gamma_k \widetilde{k}_t + \gamma_s\, s_t$, with $\mathbb{E}_t s_{t+1} = P s_t$

$$a_0\gamma_k^2 + a_1\gamma_k + a_2 = 0 \qquad \text{pick the } \textit{stable} \text{ root } |\gamma_k| < 1 \text{ (saddle path)}$$
$$\gamma_s = -(b_0 P + b_1)\big[(a_0\gamma_k + a_1)I + a_0 P\big]^{-1}$$

- State-space form with $X_t = [\widetilde{k}_t, s_t, 1]'$ and observables $Y_t = [\widetilde{y}_t, \widetilde{x}_t, \widetilde{l}_t, \widetilde{g}_t]'$:

$$X_{t+1} = \mathcal{A}X_t + \mathcal{B}\varepsilon_{t+1}, \qquad Y_t = \mathcal{C}X_t$$

  - likelihood computed with the Kalman filter $\Rightarrow$ MLE of (P_0, P, Q)
  - In practice: Blanchard–Kahn / Klein (2000) / Sims' `gensys`, or Dynare
  - Great Depression: shocks too large $\Rightarrow$ non-linear finite elements + 459-state Markov chain

Results

Parametrization

- Standard functional forms: $F(k,l) = k^{\alpha} l^{1-\alpha}$, $U(c,l) = \log c + \psi \log(1-l)$
- Parameters (annualized): $\alpha = 0.35$, $\psi = 2.24$, depreciation 4.64%, time preference 3%, population growth 1.5%, technology growth 1.6%
- Data adjustments: remove sales taxes, add service flow of consumer durables (durables purchases = investment), remove military compensation (pre-WWII)
- Vector AR(1) for wedges estimated by MLE separately for:
  - annual data 1901–1940 $\Rightarrow$ Great Depression (non-linear decision rules)
  - quarterly data 1959:1–2004:3 $\Rightarrow$ 1982 recession and postwar (log-linear)
- Postwar estimates: very persistent wedges, diagonal of $P \approx (0.98, 0.96, 0.90, 0.97)$ for (A, τ_l, τ_x, g)
- ***Our replication***: same calibration and wedge process, US quarterly data 1948:1–2026:3 (FRED: real GDP, private investment, government consumption, hours per capita)

Postwar US, 1948–2026

	Wedges		Output components	
	$\sigma(\cdot)/\sigma(y)$	$\rho(\cdot, y)$	$\sigma(y^i)/\sigma(y)$	$\rho(y^i, y)$
Efficiency	0.53 (0.63)	0.50 (0.85)	0.41 (0.45)	0.50 (0.85)
Labor	1.78 (0.92)	0.70 (0.71)	1.04 (0.65)	0.68 (0.70)
Investment	1.03 (1.16)	−0.10 (0.47)	0.53 (0.57)	−0.01 (0.52)
Government consumption	1.91 (1.51)	0.10 (−0.33)	0.73 (0.68)	0.08 (−0.36)

Table: HP(1600)-filtered quarterly US data, 1948:1–2026:3, CKM calibration and wedge process.
In gray: same code on CKM's data, 1959:1–2004:3. y^i: output in the model with wedge i alone

- Labor wedge: most volatile, strongly procyclical; its output component tracks the data
- Efficiency wedge: still procyclical, but much less than in 1959–2004 (0.50 vs. 0.85)
- Investment wedge: acyclical, output component uncorrelated with y: ***"tertiary role"***
- Government wedge (here without net exports): acyclical
- Similar without COVID (1948–2019): $\rho(\cdot, y) = 0.58,\ 0.66,\ 0.04,\ 0.14$

A simple replication: variance decomposition, 1948–2026

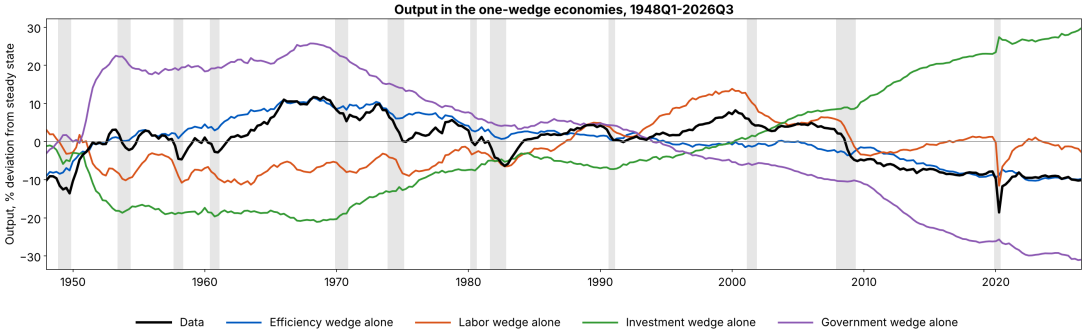
- One-wedge economies: wedge i at its measured path, others at steady state; linear rules $\Rightarrow v_t = \sum_i v_t^i$
- Share of wedge i in the variance of $v \in \{y, l, x, c\}$: $\mathrm{cov}(v^i, v)/\mathrm{var}(v)$, HP-filtered (sums to 100)

	% of variance due to			
Variable	Efficiency	Labor	Government	Investment
Output	22.0	74.2	−5.1	8.9
Labor hours	−0.8	120.1	−17.8	−1.4
Investment	0.3	70.6	−44.0	73.1
Consumption	18.7	52.2	32.5	−3.4

$\Rightarrow$ Labor $\approx 3/4$, efficiency $\approx 1/5$ of output variance; labor wedge $\approx$ all of hours

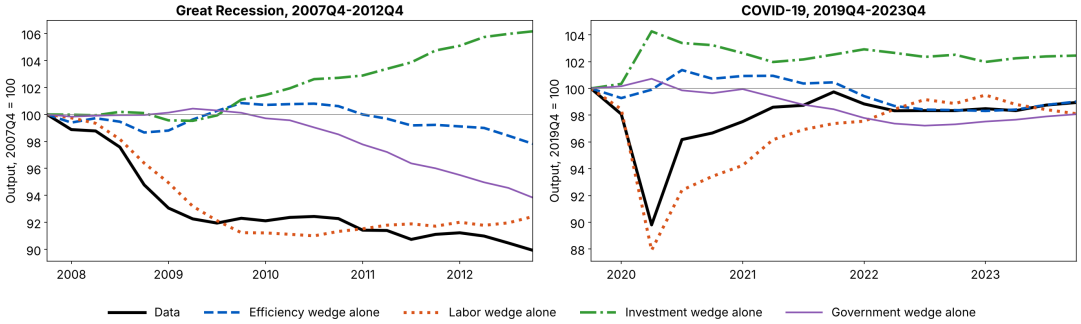
- Investment wedge matters for investment only, offset by the government wedge
- Sims (2024), 1947–2016, independent AR(1)s: efficiency $\approx 60\%$, labor $\approx 40\%$
  $\Rightarrow$ the split depends on the wedge process (correlated innovations in CKM's VAR)

Output in the one-wedge economies, 1948–2026



- Wedge *i* at its measured path, others at steady state: lines sum to data (up to linearization)
- Labor: 1990s boom, 2001, 2008–09, 2020; efficiency: 1960s boom, post-2005 slowdown
- Investment and government wedges: low-frequency trends that largely offset each other

The Great Recession and COVID-19



Output relative to trend, $t_0 = 100$. Each economy starts from the measured capital stock at t_0; wedge i follows its measured path, the other wedges stay at their t_0 values.

The Great Recession and COVID-19: accounting

	Great Recession	**COVID-19**	
Change in log output from t_0, %	2007:4–2012:4	2019:4–2020:2	2019:4–2023:4
Data	-10.6	-10.8	-1.1
Efficiency wedge alone	-2.2	-0.1	-1.0
Labor wedge alone	-7.9	-12.9	-1.9
Investment wedge alone	$+6.0$	$+4.2$	$+2.4$
Government wedge alone	-6.4	$+0.7$	-2.0

- ***Great Recession***: a labor wedge recession (as in BCKM, 2016): $\approx 3/4$ of the fall in output and -7.9 of the -9.5% fall in hours; efficiency wedge: slow decline after 2010
- Investment wedge goes the "wrong" way: τ_x falls by ≈ 20 pp
- ***COVID-19***: an extreme labor wedge recession: τ_l up ≈ 16 pp in 2020:2 (lockdowns act as a tax on work); efficiency wedge flat; output back near trend by 2023

Interpreting the results and critiques

- ***The labor wedge***: failure of the static labor FOC is a robust fact
  - Hall (1997), Galí, Gertler & López-Salido (2007), Shimer (2009), Karabarbounis (2014)
  - interpretations: countercyclical markups (sticky prices), wage markups, search frictions, labor supply "shocks"
  - CKM (2009): Smets–Wouters' wage-markup shock $\approx$ labor wedge $\Rightarrow$ structural?
- ***Financial frictions*** are *not* ruled out: they can appear as efficiency wedges (misallocation) or labor wedges (working capital), not only as investment wedges
- ***Critiques***: Christiano & Davis (2006): results sensitive to the specification of the investment wedge (capital income vs. investment tax) and to the correlation of wedges (spillovers)
- ***Vs. SVARs***: decomposition by *wedges*, no need to "name" primitive shocks ($R\eta_t = \varepsilon_t$): the equivalence results play the role of identifying assumptions

Conclusion

Conclusion

- Business cycles: comovements of output, consumption, investment, hours with robust patterns of volatility and persistence, in the US and in Europe

- Business Cycle Accounting:
  - prototype growth model + four wedges; equivalence with detailed models with frictions
  - measure wedges, feed them back one at a time
  - numerically: log-linearization around the steady state + Kalman filter for the VAR of wedges

- Main results:
  - US: efficiency and labor wedges account for most fluctuations; investment wedge *tertiary*
  - Great Recession: labor wedge in the US, efficiency wedge in most of Europe, investment wedge in Spain/Ireland

$\Rightarrow$ Next lectures: models that generate these wedges:
nominal rigidities (NK), firm heterogeneity and misallocation, HANK

Appendices

The cost of business cycles

- Consumption $C = A\, e^{-\sigma^2/2}\varepsilon$ with $\log \varepsilon \sim \mathcal{N}(0, \sigma^2)$, so that $\mathbb{E}C = A$. CRRA utility $u(C) = \frac{C^{1-\gamma}}{1-\gamma}$
- How much λ would you pay to eliminate fluctuations of C around $\mathbb{E}C$?

$$\mathbb{E}\, u\big((1+\lambda)C\big) = u(A) \qquad \Rightarrow \qquad \log(1+\lambda) = \frac{\gamma}{2}\sigma^2 \quad \Rightarrow \quad \lambda \approx \frac{\gamma}{2}\sigma^2$$

- s.d. of log real per capita consumption around trend in the US: $\sigma \approx 0.032$
- Risk aversion: macro and micro estimates around $\gamma \approx 1$, at most ≈ 5
- For $\gamma = 1$: $\lambda \approx 0.0005$: consumers would give up only 0.05% of consumption to eliminate business cycles

$\Rightarrow$ Small because aggregate volatility is small; much larger with incomplete markets and uninsurable idiosyncratic risk (Krusell & Smith 1999, Krusell, Mukoyama, Sahin & Smith 2009)

Equivalence III: BGG financial frictions → investment wedge

- Entrepreneurs with net worth n_t produce capital, idiosyncratic shock ω privately observed, costly state verification (monitoring cost μ), debt-like contract with cutoff $\bar{\omega}_t$
- Prototype economy with capital income tax τ_{kt} such that

$$\left[1-\tau_{kt}\right]\left[A_t F^*_{kt}-\delta\right]+1=\left[\Gamma(\bar{\omega}^*_t)-\mu G(\bar{\omega}^*_t)\right]R^*_{kt}+\frac{n^*_{t-1}}{k^*_{t-1}}\frac{U^*_{ct-1}}{\beta\pi(s^t|s^{t-1})U^*_{ct}}$$

  and government consumption $g_t = c^*_{et} + \mu G(\bar{\omega}^*_t) R^*_{kt} k^*_{t-1}$ (entrepreneurs' consumption + monitoring costs)
- ***Proposition 6***: aggregate allocations coincide
- Open economy (Proposition 7): g_{it} = net exports $F(k^*_{it}, l^*_{it}) - x^*_{it} - c^*_{it}$